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Abstract

Automated Diagnostic Hypotheses from 3D chest Computed Tomography requires clinically interpretable methods to support radiological decisionmaking. Current State-of-the-Art deep learning models often rely on posthoc explainability techniques that provide approximate localizations and lack direct traceability to the models decisions. Furthermore, SotA models rarely support simultaneous multiclass classification and grounded localization, leading to a fragmented approach to diagnosis. This also hinders the ability to benchmark models across multiple dimensions of performance, including classification accuracy, segmentation quality, and explainability. This study aims to address these challenges through two contributions: First, we introduce an inherently interpretable dictionary learning framework in which detected abnormalities can be backprojected to the original voxel space, allowing diagnostic decisions to be traced back to relevant anatomical structures. Second, we develop a unified evaluation framework to benchmark SotA models across three axes: (i) classification accuracy, (ii) segmentation quality, and (iii) explainability, using standardized metrics and a publically available 3D chest CT dataset with expertly annotated abnormality localizations. Our method is benchmarked against SotA models using this framework, demonstrating competitive classification and segmentation performance while providing improved interpretability compared to deep learning baselines.

Acknowledgments

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Preface

This thesis is submitted.........

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Acronyms

CXR	-	Chest X-Ray
CT	-	Computed Tomography
CAD	-	Computer-Aided Diagnosis
CNN	-	Convolutional Neural Network
ViT	-	Vision Transformer
XAI	-	Explainable Artificial Intelligence
GGO	-	Ground-Glass Opacity
COPD	-	Chronic Obstructive Pulmonary Disease

Chapter 1

Introduction

Developing a unified pipeline for multi-class abnormality classification and grounded localization addresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is accountable and its claims are supported by clinical evidence.

1.1 Background

Lung disease remains a leading cause of global morbidity and mortality []. Research shows that early detection of pulmonary diseases significantly improves patient outcomes. For this, radiological imaging tools such as Chest X-Rays (CXR) and Computed Tomography (CT), plays an important role. However, the increasing volume of such imaging data increases radiologists' workload, leading to delays and potential errors in diagnosis. This has motivated the development of automated Computer-Aided Diagnosis (CAD) systems, which use computational algorithms and Artificial Intelligence (AI) to assist in tasks such as abnormality detection, classification, and segmentation [1].

Our research mainly focuses on the development of a CAD system using 3 Dimensional (3D) CT data. While CXR is commonly used as the first imaging test due to its low cost and low radiation exposure, studies show that their sensitivity to abnormailities is limited [2]. Comparatively, CT scans produce high resolution cross-sectional slices that helps detect smaller, early-stage, and subtle lung pathologies that are frequently missed by standard CXR.

However, CT analysis is complex and time-consuming, requiring radiologists to review hundreds of slices per scan, which are often inconsistent due to artifacts, anatomical variations and different scanners/protocols. Furthermore, different anatomical structures can share identical grayscale values. Telling them apart requires expert clinical judgment which is often subjective. High sensitivity can lead to detection of clinically insignificant abnormalities, while visual fatigue can lead to missed findings [3].

To address these challenges, recent research and clinical deployments of AI-based CAD systems are fundamentally changing how radiologists approach CT analysis. AI models such as 3D Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have shown to reduce interpretation time, and improve lesion detection sensitivity compared to traditional manual methods. Many of these high performing models are Deep Learning based, which are criticized for their black box nature, where their internal reasoning is difficult to interpret. This lack of interpretability is a significant barrier for clinical adoption, as radiologists require evidence to support a model's diagnostic claims.

This has motivated the development of explainable AI (XAI) methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) that helps to visualize exactly which parts of an image a model focuses on when making a specific prediction. Such XAI methods are highly valuable for building clinical trust in deep learning models. However, they often lack the fine-grained, pixel-level precision required to identify tiny lesions, sometimes highlighting incorrect or confounding features [4].

Recent advances in sparse representation models such as Sparse Autoencoders and Dictionary Learning have shown to provide inherent interpretability in medical imaging. This stems from their ability to reconstruct complex, high-dimensional data as a linear combination of ”atoms” or features. This allows the use of efficient algorithms to easily map the output sparse domain back to the original input feature space, providing a linear relationship between the model's decision and anatomical structures. Our work builds on these advances by applying dictionary learning to 3D Chest CT data for multi-class abnormality classification and grounded localization.

1.2 Problem Statement

The clinical utility of Deep Learning based approaches is hindered by a lack of interpretability. While tools like Grad-CAM provide post-hoc explanations to model decisions, these are often insufficient for medical professionals who require a granular understanding of the model's decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into clinical workflows, as it lacks the accountability necessary for legal and ethical compliance.

1.3 Objectives and Scope

The primary goal of this study is to address the limitations of existing AI-based approaches for abnormality classification and localization in chest CTs, by developing an inherently interpretable dictionary learning framework. This approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facilitates validation and adoption in radiological workflows. The specific objectives are as follows:

1.3.1 Objectives

- Implement a Dictionary Learning based CAD system using a publicly available annotated 3D chest CT dataset.

- Map dictionary atoms back to anatomical features and visualize their spatial contribution maps.

- Develop a unified evaluation framework for assessing the performance and interpretability of the proposed approach.
- Benchmark model performance against SotA classification and segmentation models on the same dataset.

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning based approach for classification and grounded localization of lung abnormalities using volumetric 3D CT data, with the following limitations:

- This study is limited to 2 thoracic abnormalities: Pulmonary Nodules, and Ground-Glass Opacities, based on expert radiologist recommendation and dataset availbility.
- This study will utilize a single publicly available dataset for training and evaluation, which may limit generalizability to other populations and scanner types.
- The proposed approach will be limited only to interpretable methods, and will not explore approaches that provide better performance for less explainability.

Chapter 2

Literature Review

2.1 Clinical Background

Chronic respiratory diseases account for more than 4 million deaths globally per year []. Research shows that early detection of pulmonary diseases, particularly lung cancer and Chronic Obstructive Pulmonary Disease (COPD) dramatically lowers mortality rates. Two landmark studies, the National Lung Screening Trial (NLST) and the Nederlands-Leuvens Longkanker Screenings Onderzoek (NELSON) trial, have shown that low-dose CT screening educes lung cancer mortality by 20% to 33% compared to CXR [5, 6].

2.1.1 GGO and Pulmonary Nodules

What are pulmonary nodules and ground-glass opacities (GGOs)? Why are they clinically important? Why is early detection essential? Why are these abnormalities difficult to detect?

2.1.2 Computed Tomography (CT)

What role does CT play compared to other imaging modalities? What challenges exist in interpreting 3D chest CT scans?

2.2 Computer-Aided Diagnosis (CAD) Systems

What is Computer-Aided Diagnosis? What tasks does CAD perform? * classification * detection * localization * segmentation

2.2.1 Evolution of CAD Systems

How have CAD systems evolved? Why has deep learning become dominant?

2.3 Deep Learning Approaches

General idea on CNNs, ViTs, and other deep learning architectures.

How well do they perform? Which datasets are commonly used? What evaluation metrics are reported?

2.3.1 nnU-Net

2.3.2 BioMedParse

2.3.3 TotalSegmentor

2.3.4 Merlin

2.3.5 MedSAM

2.3.6 CT-CLIP

2.4 Explainable AI (XAI) in Medical Imaging

Why is interpretability important in healthcare? What regulatory and ethical requirements exist? Why do clinicians require explanations? Why is explainability especially important for CT diagnosis?

2.4.1 SHAP

2.4.2 LIME

2.4.3 Grad-CAM

Also add Grad-CAM++

2.4.4 Other CAM Methods

Hires-CAM, Score-CAM, XGrad-CAM, etc.

2.4.5 Limitations of XAI Methods

2.5 Sparse Representation

Why is sparsity useful? How is sparse representation learned?

2.5.1 Sparse Coding

What is sparse coding? What is overcomplete representation? Why is sparse representation robust?

Two types: Greedy Algorithms and Convex Relaxation

2.5.2 Greedy Algorithms

Matching Pursuit (MP)

Orthogonal Matching Pursuit (OMP)

2.5.3 Convex Relaxation Algorithms

Least Angle Regression (LARS)

Fast Iterative Shrinkage-Thresholding Algorithm (FISTA)

2.6 Dictionary Learning

What is a dictionary atom? How is reconstruction performed? Why are atoms interpretable? What optimization algorithms exist?

2.6.1 K-Means Singular Value Decomposition (K-SVD)

2.6.2 Method of Optimized Directions (MOD)

2.6.3 Online Dictionary Learning (ODL)

2.7 Discriminative Dictionary Learning

Why isn't reconstruction enough? What is discriminative dictionary learning?

2.7.1 Label-Consistent K-SVD (LC-KSVD)

2.7.2 Fischer Discriminative Dictionary Learning (FDDL)

2.7.3 Frozen Dictionary Learning

2.8 Dictionary Learning in Medical Imaging

2.8.1 Image Reconstruction

2.8.2 Classification and Segmentation

2.9 Evaluation Metrics

2.9.1 Classification

Accuracy Precision Recall F1 ROC-AUC

2.9.2 Localization and Segmentation

Dice IoU Hausdorff Sensitivity

2.9.3 Interpretability

Faithfulness Localization accuracy Sparsity Clinical plausibility

2.10 Research Gaps

Chapter 3

Methodology

Chapter 4

Conclusion

Appendix A

Essential English Grammar - A Handout

The Appendices are numbered as A, B,.. and so on. Sections and Subsections of the appendix then would be A.1, A.1.1, A.2 etc.

A.1 Simple Present Tense

A.2 Perfect Tense

A.3 Past Past Perfect Tense

A.4 Passive Voice

A.5 Technical Writing Fundamentals

A.6 Proof Reading

Appendix B

Graphical Interpretation of Data

The graphical interpretation given below shows the relationship between x and y is acceptable to the purpose of the project

B.1 Data Analysis

B.1.1 Statistical Analysis

B.1.2 Empirical Analysis

B.2 The X vs Y vs Z

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